STATS 7022 - Data Science Assignment 3: Question 2

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(a)

Use the formula of integration by parts,

$$\int u\,dv = uv - \int v\,du.$$

Consider $u = g''(x), dv = h''(x)$, so $v = h'(x)$, and makes the following substitution,

$$\int_a^b g''(x)h''(x) = g''(x)h'(x)\big|_a^b - \int_a^b g'''(x)h'(x)dx \tag{1}$$

As $g(x)$ is a natural cubic spline with n knots, we could break the integral and interval in (2) into n parts, and we have

$$\int_a^b = \int_a^{x_1} + \sum_{j=1}^{n-1}\left(\int_{x_{j+1}}^{x_j}\right) + \int_{x_n}^b. \tag{2}$$

Hence substituting (2) into (1), we have

$$\begin{aligned}\int_a^b g''(x)h''(x) =& g''(x)h'(x)\big|_a^{x_1} + \sum_{j=1}^{n-1}\left(g''(x)h'(x)\big|_{x_j}^{x_{j+1}}\right) + g''(x)h'(x)\big|_{x_n}^b \\ &- \int_a^{x_1} g'''(x)h'(x)dx - \sum_{j=1}^{n-1}\left(\int_{x_{j+1}}^{x_j} g'''(x)h'(x)dx\right) - \int_{x_n}^b g'''(x)h'(x)dx. \end{aligned} \tag{3}$$

Q.E.D.

(b)

To show that

$$g''(x)h'(x)\big|_a^{x_1} + \sum_{j=1}^{n-1}\left(g''(x)h'(x)\big|_{x_j}^{x_{j+1}}\right) + g''(x)h'(x)\big|_{x_n}^b = 0. \tag{4}$$

Recall that $g(x)$ is a natural cubic spline, we have $g''(x) = 0$ at the endpoints a and b.

Expand the left-hand side in (4),

$$
\begin{aligned}
& g''(x)h'(x)\Big|_a^{x_1} + \sum_{j=1}^{n-1}\left(g''(x)h'(x)\Big|_{x_j}^{x_{j+1}}\right) + g''(x)h'(x)\Big|_{x_n}^{b} \\
=& g''(x_1)h'(x_1) - g''(a)h'(a) + \sum_{j=1}^{n-1}\left(g''(x_j)h'(x_j) - g''(x_{j+1})h'(x_{j+1})\right) + g''(b)h'(b) \\
=& \cancel{g''(x_1)h'(x_1)} - g''(a)h'(a) + \cancel{g''(x_2)h'(x_2)} - \cancel{g''(x_1)h'(x_1)} + \cdots \\
& + \cancel{g''(x_n)h'(x_n)} - \cancel{g''(x_{n-1})h'(x_{n-1})} + g''(b)h'(b) - \cancel{g''(x_n)h'(x_n)} \\
=& -g''(a)h'(a) + g''(b)h'(b) \\
=& 0
\end{aligned}
$$

Q.E.D.

(c)

To show that

$$\int_a^b g'''(x)h'(x)dx = -\sum_{j=1}^{n-1}\left(\int_{x_j}^{x_{j+1}} g'''(x)h'(x)\right). \tag{5}$$

Substitute (3) into (4), we have

$$\int_a^b g'''(x)h'(x)dx = \int_a^{x_1} g'''(x)h'(x)dx - \sum_{j=1}^{n-1}\left(\int_{x_{j+1}}^{x_j} g'''(x)h'(x)dx\right) - \int_{x_n}^{b} g'''(x)h'(x)dx$$

Using integral by parts on each interval$[x_j, x_{j+1}]$, we have

$$\int_{x_j}^{x_{j+1}} g'''(x)h'(x)dx = g''(x)h'(x)\Big|_{x_j}^{x_{j+1}} - \int_{x_j}^{x_{j+1}} g''(x)h''(x)dx, \tag{6}$$

summing (6) from $j = 1$ to $n-1$, we have

$$\int_a^b g'''(x)h'(x)dx = \sum_{j=1}^{n-1}\left[g''(x_{j+1})h'(x_{j+1}) - g''(x_j)h'(x_j)\right] - \sum_{j=1}^{n-1}\left(\int_{x_j}^{x_{j+1}} g'''(x)h'(x)\right),$$

as we did in (b), we can cancel the terms in the summation, and we have

$$\int_a^b g'''(x)h'(x)dx = -\sum_{j=1}^{n-1}\left(\int_{x_j}^{x_{j+1}} g'''(x)h'(x)\right).$$

Q.E.D.

(d)

Since $g(x)$ is a natural cubic spline, we have $g'''(x)$ is a constant, which could be denoted as c_j on each interval $[x_j, x_j + 1]$. Then we have

$$\int_{x_j}^{x_{j+1}} g'''(x)h'(x)dx = c_j\int_{x_j}^{x_{j+1}} h'(x)dx = c_j\left[h(x_{j+1}) - h(x_j)\right]. \tag{7}$$

Hence $h(x) = \tilde{g}(x) - g(x)$, and $\tilde{g}(x_i) = g(x_i) = y_i$, we have $h(x_i) = 0,\ i = 1, \cdots, n$. So (7) could be rewritten as

$$\int_{x_j}^{x_{j+1}} g'''(x)h'(x)dx = c_j\left[h(x_{j+1}) - h(x_j)\right] = c_j(0-0) = 0.$$

Thus we have

$$\int_a^b g'''(x)h'(x)dx = -\sum_{j=1}^{n-1}\left(c_j\left[h(x_{j+1}) - h(x_j)\right]\right) = 0.$$

Q.E.D.

(e)

Since $\tilde{g}(x) = g(x) + h(x)$, we have $\tilde{g}''(x) = g''(x) + h''(x)$.

And we could also have

$$\int_a^b \tilde{g}''(x)^2 dx = \int_a^b g''(x)^2 dx + \int_a^b h''(x)^2 dx + 2\int_a^b g''(x)h''(x)dx.$$

As we have shown before,

$$\int_a^b g''(x)h''(x)dx = g''(x)h'(x)\big|_a^b - \int_a^b g'''(x)h'(x)dx = 0 - 0 = 0.$$

Thus we have

$$\int_a^b \tilde{g}''(x)^2 dx = \int_a^b g''(x)^2 dx + \int_a^b h''(x)^2 dx. \tag{8}$$

As $\int_a^b h''(x)^2 dx$ must be non-negative, we have

$$\int_a^b \tilde{g}''(x)^2 dx \geq \int_a^b g''(x)^2 dx.$$

Q.E.D.

(f)

From (8), we have

$$\int_a^b \tilde{g}''(x)^2 dx = \int_a^b g''(x)^2 dx + \int_a^b h''(x)^2 dx.$$

- **Sufficiency**

  If $h(x) = 0$, then accordingt to (8) we could obtain

  $$\int_a^b \tilde{g}''(x)^2 dx = \int_a^b g''(x)^2 dx + 0 = \int_a^b g''(x)^2 dx.$$

- **Necessity**

  If $\int_a^b \tilde{g}''(x)^2 dx = \int_a^b g''(x)^2 dx$, then according to (8),

  $$\int_a^b h''(x)^2 dx = 0.$$

  Hence $h''(x)^2 \geq 0$, then $h''(x) = 0$ for all $a < x < b$.

  Thus $h(x)$ is a linear function, as we have $h(x_i) = 0,\ i = 1, \cdots, n$.

So $h(x)$ is a linear function with n knots, and it must be a constant function.

Hence $h(x) = 0$ for all $x \in (a, b)$.

So if and only if $h(x) = 0$for all $a < x < b$ can we have

$$\int_a^b \tilde{g}''(x)^2 dx = \int_a^b g''(x)^2 dx.$$

(g)

To show that the natural cubic spline is the unique function of

$$\hat{f}(x) = \min_f \left\{ \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int_a^b f''(x)^2 dx \right\}$$

As the property of the natural cubic spline, we have $f(x_i) = y_i$. Thus we have

$$\sum_{i=1}^n (y_i - f(x_i))^2 = 0.$$

For any other function $\tilde{f}(x)$ which is not a natural cubic spline, could not satisfy the above equation.
From (e) and (f), for any other function $\tilde{f}(x)$ satisfying $\tilde{f}(x_i) = y_i$, we have

$$\int_a^b \tilde{f}''(x)^2 dx \geq \int_a^b f''(x)^2 dx.$$

Where $f(x)$ is the natural cubic spline interpolating y_i at x_i.
So if and only if $\tilde{f}(x) = f(x)$, we can minimize the integral $\int_a^b \tilde{f}''(x)^2 dx$.
Thus $f(x)$ must be a natural cubic spline which has knots at $x_1, \cdots, x_n$.